

Agent Handbook — 05 Utilization Management

The payer-side wedge. Summarizes a UM request against medical-necessity criteria and runs a fairness screen — but **issuing the determination is withheld from every agent**, and an adverse recommendation is **forwarded to the medical director, never auto-denied**. Maturity: Demonstrated + Deployable-by-design; production-readiness (CSV/CSA, IdP, live connectors, pen test, HITRUST) is the engagement; AWS BAA precedes any PHI.

Workflow + human gate

Receive a UM/medical-necessity request → assemble clinical evidence → summarize against **MCG / InterQual** criteria → run the **four-fifths fairness screen** on the recommendation → **suspend at the Medical-Director gate**. An **adverse** recommendation is **forwarded** to the medical director, **never auto-denied**. **Named human gate**: Medical Director.

Systems / connectors (with standards)

- **EHR / FHIR** — Epic / Oracle Health via Amazon HealthLake (FHIR R4): clinical evidence.
- **Criteria** — **MCG / InterQual** medical-necessity criteria for grounding.
- **Payer** — **Da Vinci** (CRD/DTR/PAS); **X12 278** for the UM transaction.

Workforce roles to map in the IdP

UM_NURSE (review/assemble) and UM_MEDICAL_DIRECTOR (the determination gate). Map to custom:hpp_role.

Withheld authority (what it never does)

payer.issue_determination is WITHHELD from ALL agents. No agent issues a coverage/UM determination. An adverse recommendation is **forwarded** to the Medical Director; it is **never auto-denied**. The four-fifths fairness screen runs on every flag/rank.

Deploy (fixture mode, no PHI)

scripts/build_lambdas.sh

scripts/deploy.sh 05-utilization-management dev portable native s3://my-cfn-staging/hpp 10.34.0.0/16

- Params: infra/cloudformation/params/05-utilization-management.json · Terraform: terraform apply -var-file=envs/05-utilization-management.tfvars
- VPC CIDR 10.34.0.0/16.

Human-gate smoke test

1. Sign in as a seeded UM_NURSE; drive a UM summary with an **adverse** recommendation → confirm it is **forwarded** to the Medical Director and **suspends** at that gate (never auto-denied).
2. Confirm the **four-fifths fairness screen** ran on the recommendation.
3. As UM_MEDICAL_DIRECTOR, approve → proceeds; approval written to the **append-only PHI-masked audit**. Reject → no determination issued.
4. Attempt payer.issue_determination from any agent path → **denied by default** (withheld from all).

Key regs

HIPAA Privacy/Security, HITECH; **CMS AI-in-UM guidance** (no automated denials of medically necessary care); **CMS-0057-F**; Section 1557 (nondiscrimination — the fairness screen); plan medical policy.

Go-live checklist

- ☐ AWS BAA + SI BAA executed; no PHI before both.
- ☐ IdP federated; UM_NURSE / UM_MEDICAL_DIRECTOR mapped to custom:hpp_role.
- ☐ MCG/InterQual criteria + FHIR connectors validated; X12 278 path confirmed.
- ☐ **Four-fifths fairness screen** validated on customer data; adverse-action routing confirmed.
- ☐ CSV/CSA signed off; HITL SLA ratified; Guardrail (unauthorized-determination denied topic) confirmed.
- ☐ Pen test + HITRUST/SOC 2 evidence assembled; observability + runbooks adopted.